Quantitative Investment: An Overview

MFE5340: AI in FE: Quantitative Investment

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Spring 2026



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- Tendency for Overfitting
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- Strive for Excellence, Not Perfection
- The “Magic” Formula

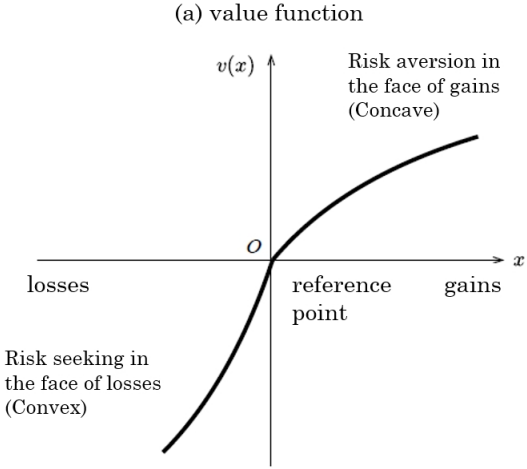


Risk Preference Cannot Explain Investor Choices

- Since 1926, the average annual real return of the U.S. stock market is 7%, while government bonds have returned less than 1%.
- The proportion of bond investors far exceeds that of stock investors, leading to the equity risk premium puzzle.
- Using the returns and standard deviations of stocks and bonds, and plugging them into a utility function, the only "reasonable" explanation for this phenomenon is that investors have an extremely high level of risk aversion.

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- Richard Thaler combined **Prospect Theory** and **Mental Accounting** to explain this phenomenon.
- Prospect Theory: Investors exhibit loss aversion, and its value function shows that the utility of losses is greater than the utility of gains. This increases the appeal of risk-free assets like bonds.
- Mental Accounting: The third part of mental accounting is about the frequency of evaluation. Investors tend to frequently check whether their stocks have profits or losses.



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Loss Aversion + Frequent Evaluation = Myopic Loss Aversion

- Generally, the pain of losses is **twice** as strong as the happiness of gains (of the same magnitude) for investors.
- Investors frequently check their stock portfolios. Gains bring only 1× happiness, while losses cause 2× pain. This frequent evaluation leads to overestimating the appeal of bonds.
- The pain from frequent evaluation causes investors to prefer bonds, which helps explain the equity premium puzzle.

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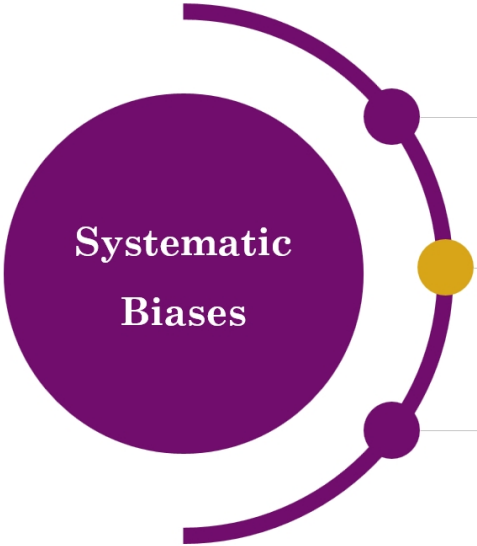
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Thinking Fast and Slow

- Everyone has two decision-making systems in their brain: the **X system** (emotion-driven) and the **C system** (logic-driven).
- When making decisions, information is first processed by the X system, which is instinctive, automatic, and effortless, but can lead to faulty decisions.
- The C system requires logical thinking to process information more thoroughly and make more rational decisions.
- Under stress, the X system becomes more prominent than the C system.



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Representativeness

When people try to determine whether data A comes from model B, or whether sample A belongs to category B, they often assess the similarity between A and B.

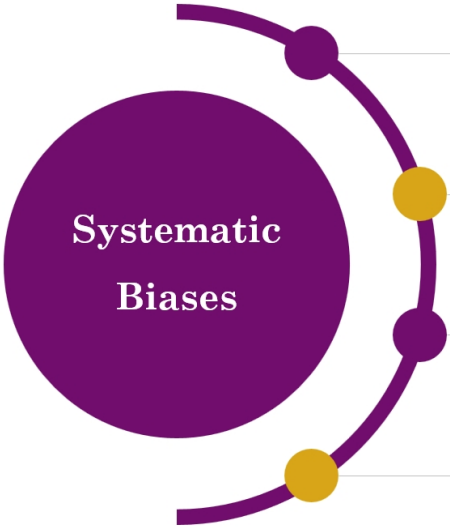
Availability

When people evaluate problems, concepts, methods, or decisions, they often rely on the examples that come to mind most easily.

Anchoring

Individuals tend to overly rely on the initial information (referred to as the *anchor*) provided to them when making decisions, even if that information is unrelated to the decision being made.

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Bandwagon Effect

People are influenced by the consensus or actions of the majority, leading them to go along with the crowd and conform to popular opinion.

Gambler's Fallacy

The incorrect belief that if something occurs more frequently than usual over a certain period, its frequency will decrease in the future (and vice versa).

Hindsight

After an event occurs, we retrospectively believe that its outcome was predictable, even though there may be no objective basis to suggest that the event was actually predictable.

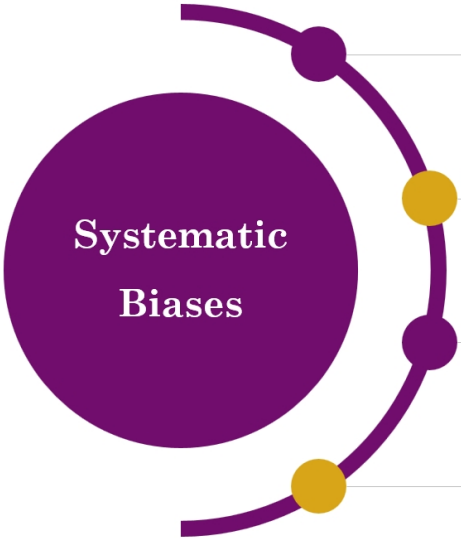
Insensitivity to Sample Size

A cognitive bias that occurs when people calculate the probability of a sample statistic without considering the sample size (the law of small numbers bias).

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Systematic Biases

2 Cognitive Bias



Outcome Bias

People often judge the quality of a decision based on its outcome, without considering the decision-making process.

Overconfidence

People's confidence in their abilities is often higher than the objective level of those abilities.

Peak-End Rule

People's evaluation of an experience largely depends on the feelings at its peak and at its end, rather than the average feeling throughout the entire experience.

Selective Perception

The tendency of individuals to perceive information in a way that aligns with their preexisting beliefs, attitudes, or expectations, and to disregard or minimize information that contradicts those beliefs.

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How to Overcome Systematic Biases?

- *You should replace humans by algorithms whenever possible, and this is really happening. Even when the algorithms don't do very well, humans do so poorly and are so noisy that just by removing the noise you can do better than people.*

  — *Daniel Kahneman, Nobel Laureate*

- *The Turtle Trading Rules*: Good traders go against human nature, and quantitative systems are the best tools to overcome human biases in trading.

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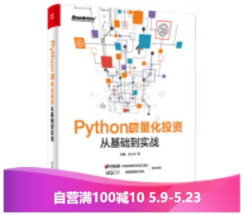
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Quantitative Investing: The Endless Stream of Books?

3 Quantitative Investment





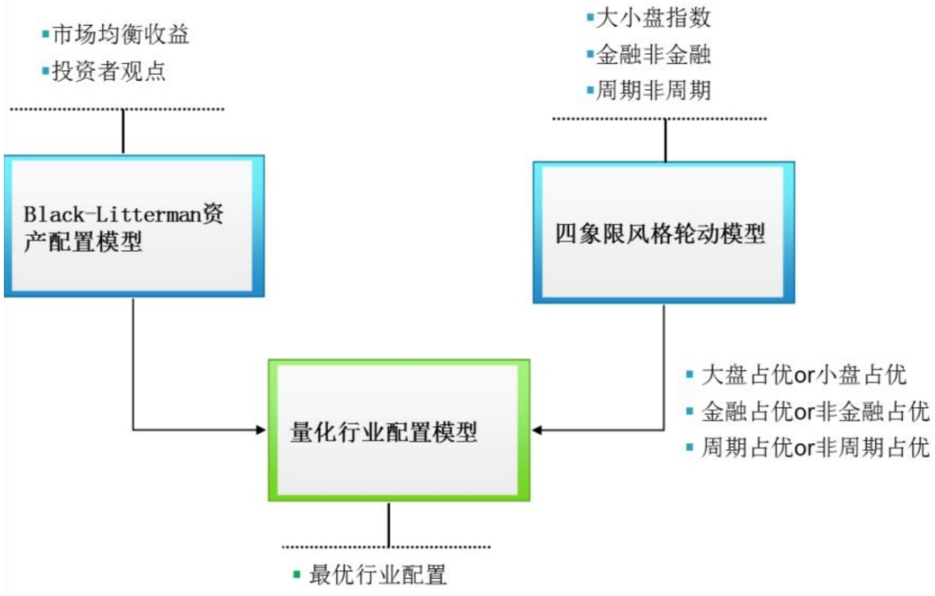








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Quantitative Investing = Quantitative Thinking + Engineered Systems

- Quantitative investing has evolved into a comprehensive system consisting of theoretical frameworks, research methodologies, and engineered systems based on modern scientific methods.
- Quantitative investing = an upgrade from experience to knowledge: The process of programmatizing the investment process is essentially a thinking process, one that abstracts and refines experience into knowledge.

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An Industrialized Production Line

- Building the entire investment process into an industrialized production line:

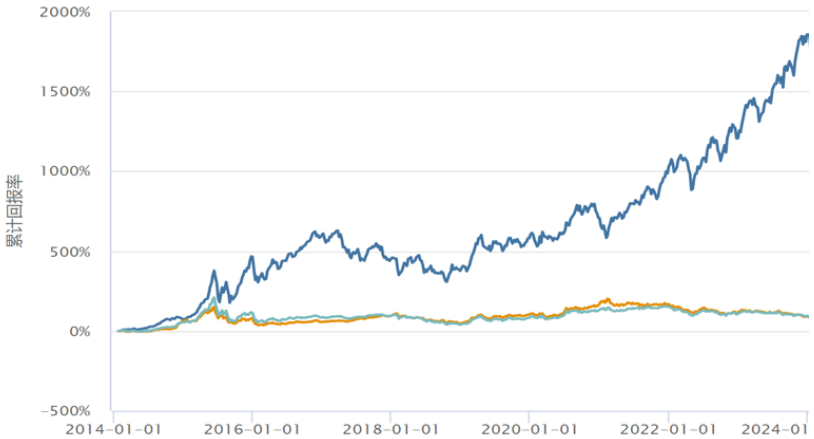


- An end-to-end approach to measure and manage **known risks**.
- Independent iteration and upgrading of each module to maintain cutting-edge research and investment practices.
- Rapid identification of potential issues during drawdowns, addressing them at their root.

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Risk Preference Bias: Ignoring Unknown Risks

- People tend to favor “high returns and low risk”.



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Risk Preference Bias: Ignoring Unknown Risks

- However, does historical data capture all risks?

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Risk Preference Bias: Ignoring Unknown Risks

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- Risk arises from the **uncertainty of variables** (known risks) and the **ambiguity of their distributions** (unknown risks).

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Type I error



Type II error

- For animals to survive, they must minimize Type II error, even if it means allowing a higher rate of Type I error (false discovery).
- This tendency has been passed down through generations during evolution. Humans tend to allow a higher rate of Type I error when analyzing problems, making overfitting more likely to occur.

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Multiple Hypothesis Testing

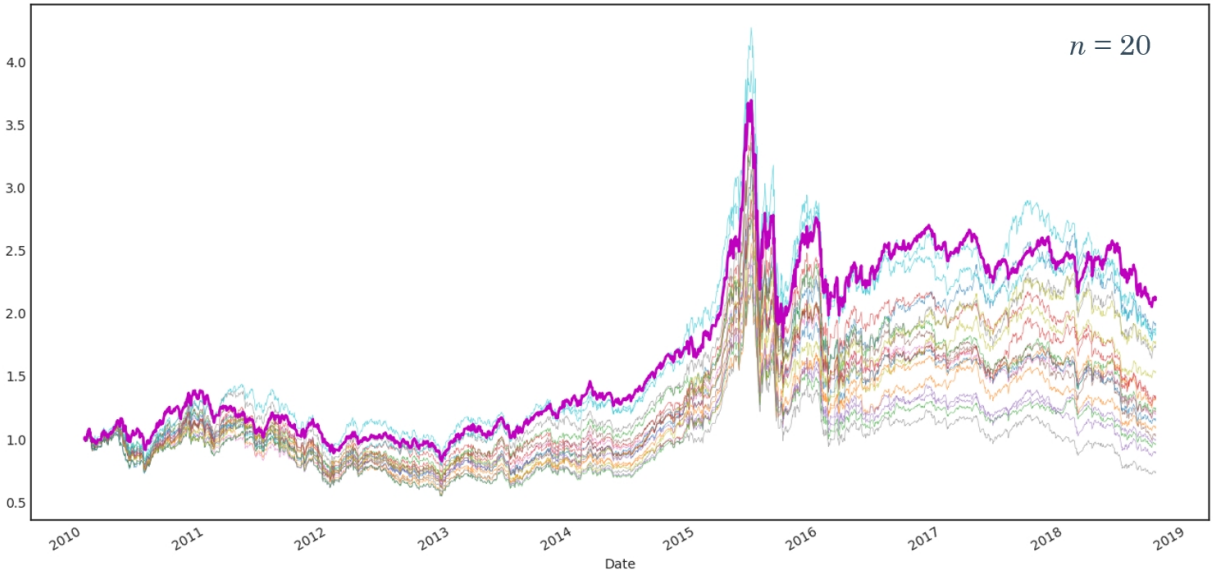
4 Tendency for Overfitting

Multiple hypothesis testing refers to the process of evaluating more than one statistical hypothesis at the same time.

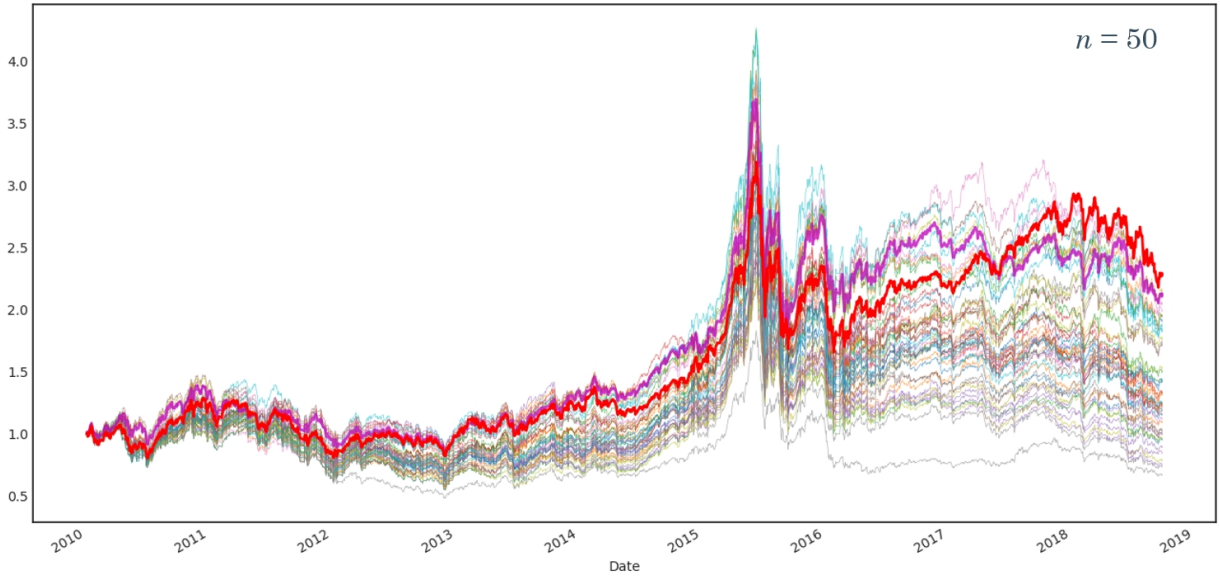
- In typical hypothesis testing, you assess whether one hypothesis is true or not based on your data.
- When you test multiple hypotheses simultaneously, the chances of making incorrect conclusions (false positives) increase. False positive appears simply due to luck (noise).

Marcos Lopez de Prado: If the researcher tries a large enough number of strategy configurations, a backtest can always be fit to any desired performance for a fixed sample length.

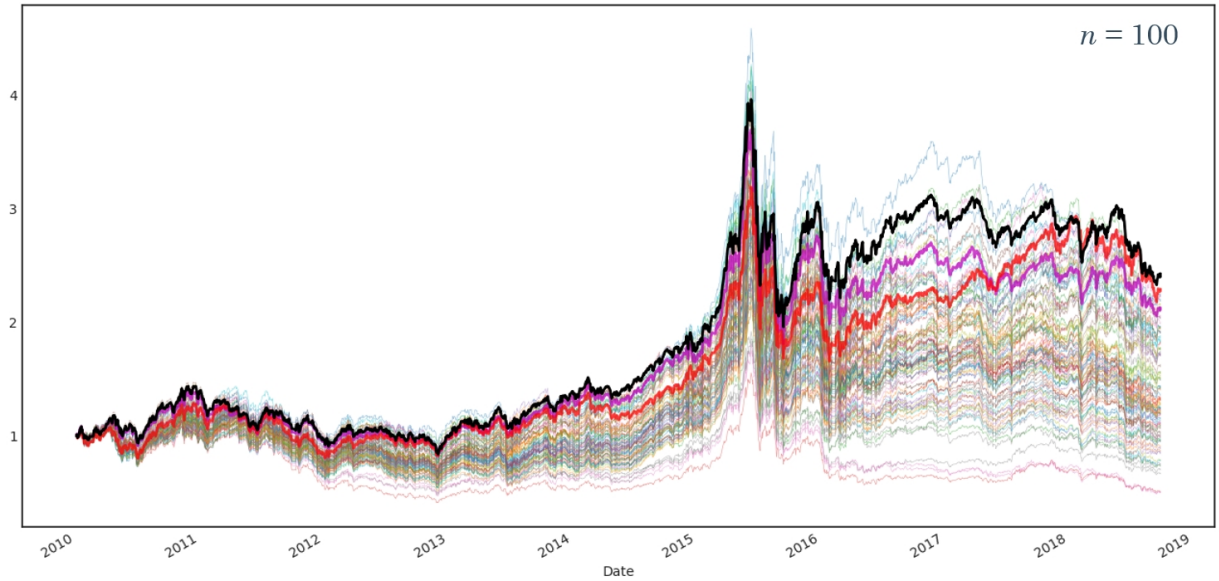
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Research conclusions depend on what kind of hypotheses are proposed, how explanatory variables are chosen, how data is processed, how data is analyzed, and how results are interpreted.

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- Should factors be extracted from the entire market, index constituents, or industries?
- Should factors be constructed using portfolio sort or cross-sectional regression (i.e., pure factors)?
- How should outliers be handled? What about normalization? What about winsorization?
- ...

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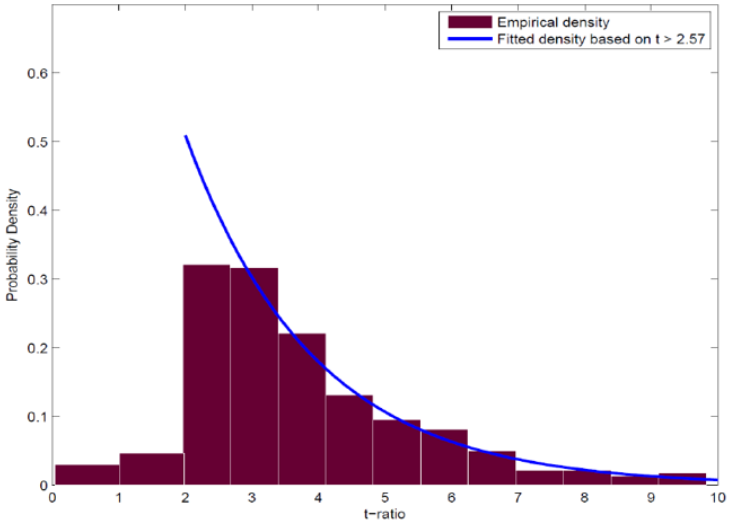
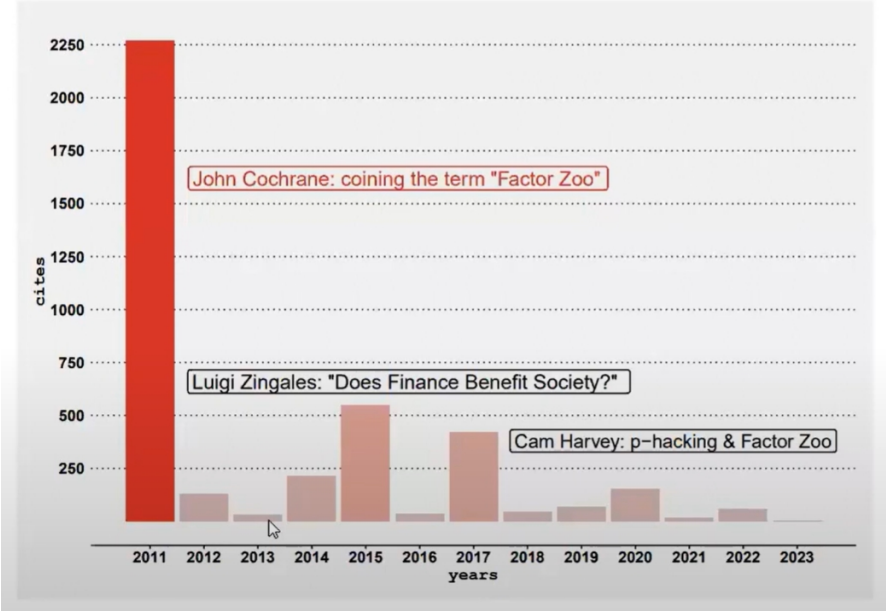


Figure 1. Distribution of reported *t*-statistics from factor studies, 1963 to 2012. Evidence from Harvey, Liu, and Zhu (2016).

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- **Frequentist methods:** This approach aims to **control Type I errors** by raising the cutoff value of t-statistics.
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- **Frequentist methods:** This approach aims to **control Type I errors** by raising the cutoff value of t-statistics.
  - Only factors with original t-statistics far exceeding the traditional cutoff of 2.0 are likely to remain significant after the adjustment.
- **Bayesian methods:** This approach aims to calculate the **posterior** probability of a factor being true. Its advantage lies in the ability to incorporate appropriate financial priors within the Bayesian framework, but its downside is the complexity of computation, often requiring simplifications.
  - Harvey and Liu (2020) proposed a double-bootstrap method, which can be considered as an extension of the Bayesian approach.

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Bryan Kelly: Finance is fundamentally different from the domains where machine learning has had success to date.

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Bryan Kelly: Finance is fundamentally different from the domains where machine learning has had success to date.

- Signal-to-noise ratio: The expected success that a human would have in a particular task.
  - Classifying images: Hand 1,000 images to a human, and ask him to categorize them as an image of a cat or not a cat, the human is gonna to achieve that with a R^2 of essentially one. (This is a high signal-to-noise environment – the signal (i.e., the cat) dominates the noise (i.e., the background) in the image.)

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  - Let's contract that with finance: Ask 1,000 experts to predict S&P500's movement direction tomorrow, the hit rate would be almost 50%. (This is just a statement about the nature of financial markets.)

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Monthly out-of-sample stock-level prediction performance (percentage R^2_{OOS})

	OLS +H	OLS-3 +H	PLS	PCR	ENet +H	GLM +H	RF	GBRT +H	NN1	NN2	NN3	NN4	NN5
All	−3.46	0.16	0.27	0.26	0.11	0.19	0.33	0.34	0.33	0.39	0.40	0.39	0.36
Top 1,000	−11.28	0.31	−0.14	0.06	0.25	0.14	0.63	0.52	0.49	0.62	0.70	0.67	0.64
Bottom 1,000	−1.30	0.17	0.42	0.34	0.20	0.30	0.35	0.32	0.38	0.46	0.45	0.47	0.42

- Source: Gu, Kelly and Xiu (2020)

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- Source: Gu, Kelly and Xiu (2020)

Bryan Kelly: Financial markets are competitative and because of that, they are extremely efficient. It's the very efficiency of markets that make them unpredictable. The signal-to-noise ratio in financial markets is extremely low.

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- Classifying images: When you go to recognize a cat, it doesn't make future cat image blurrier. We can assume that **the data generating process remains unchanged regardless of the model's learning process.**

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- Let's contract that with finance (again): In financial markets, when you trade using private information with predictive value, your profits come at the cost of **eroding** the very predictability you exploited. **As you capitalize on this information, your actions help incorporate it into market prices, making the market more efficient and at the same time less predictable.**

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Long story short, finance is just a very different system than the places where machine learning has been historically very successful.

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Typical Machine Learning vs. Finance

	Typical Problem	**Quantitative Investing**
signal-to-noise	high	very low
data dimension	many predictors many observations	many predictors few observations
prediction error	statistical issue	impact portfolio risk
overfitting	training/validation set, CV	p-hacking, investor learning
structural changes	stationary	not stationary

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						CV portfolio return r_p		
Method	Scaling	CV criterion	γ (i)	IS R^2 (ii)	CV R^2 (iii)	Mean (iv)	S.D. (v)	Sharpe Ratio (vi)
OLS	Equal	n/a	0	5.22	−1.18	4.12	11.60	0.35
Ridge	Equal	R^2	2.25	2.63	0.84	4.20	13.85	0.30
Ridge	Unequal	R^2	1.40	2.69	1.18	4.55	12.47	0.37
Ridge	Unequal	$E[r_p]$	3.11	1.75	0.89	4.58	12.94	0.35
Lasso	Unequal	R^2	0.00028	3.55	0.84	4.25	11.79	0.36

- Source: Nagel (2021)

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Mullainathan and Spiess (2016)

Estimation vs Prediction

Estimation	**Prediction**
• Strict assumptions about data generating process	• Allow for flexible functional forms
• Back out parameters	• Get predictions
• Low dimensional	• Does not adjudicate between **observably** similar functions (variables)
$\hat{\beta}$	$\hat{y}_i$

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- **High-Frequency Data Analysis:** Pattern recognition in order flow using machine learning techniques.
- **Machine Learning and Asset Pricing:** Leveraging machine learning to predict asset expected returns, enhancing traditional factor models.
- **Machine Learning for Company Fundamentals:** Machine learning applied to forecast the fundamentals of listed companies.
  - E.g., look-ahead factor models.

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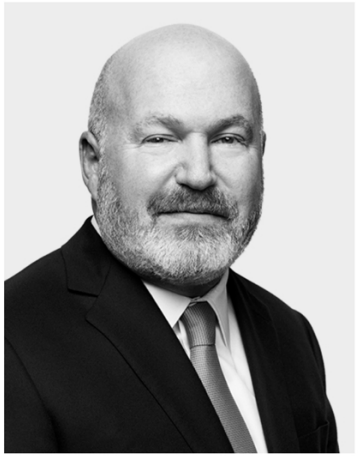
Greatest Enemy in Investing: Psychology

6 Strive for Excellence, Not Perfection

- **Even the most precise backtests cannot replicate the psychological pressure faced during live trading.**



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Cliff Asness



John M. Liew

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- **Liew: Cliff, you've seen the evidence, out of sample since the original work across time, asset classes, and geographies, that what we do has worked long time, right?**

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- **Asness: Yes, John.**

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- **Liew: And Cliff, we don't do anything we don't understand, right? I mean, we think there is economics, both classical and behavioral, and common sense behind what we do, right?**

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- **Asness: Yes, John. I've actually done much of that myself.**

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- **Liew: And Cliff, you, and the rest of us, have done this with an open mind, done it pretty obsessively, and keep concluding there is no story that holds up for the investing world being that different going forward than in the long-term past, right?**

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- **Asness: Yes, yes, John, I know.**

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- **Liew: And Cliff, you do believe that the twenty-six years we've put into working on this live, and thirty years of studying it, have led to an excellent implementation of multi-factor quant, not one that 'taste lousy', right?**

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- **Liew: Then Cliff, why are you so upset today, and so many other days, and why do you entertain doubts that you know you have zero evidence to back up – quite the opposite actually?**

- **Asness: John, I agree everything you say, but why are you not upset and how do you not go to the same dark place!!!?**

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- Advice 1: Everyone is affected by cognitive biases, and it is completely normal to have moments of frustration or temporary setbacks (Quant requires regular mental self-care). However, **never let personal willpower override the strategy.**

> *We slavishly follow the model. You do whatever it says no matter how smart or dumb you think it is.* — James Simons, Renaissance Technologies

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- **Every Asness needs his own Liew.**

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No Perfect Strategy

6 Strive for Excellence, Not Perfection

- Every strategy is based on a certain assumption about the market. The market is full of uncertainty, and at times, it will deviate from this assumption, leading to losses for the trading system.
- A good quantitative strategy is one that can be profitable **over the long term**, rather than one that makes money on every trade.

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No Perfect Strategy

6 Strive for Excellence, Not Perfection

- Every strategy is based on a certain assumption about the market. The market is full of uncertainty, and at times, it will deviate from this assumption, leading to losses for the trading system.
- A good quantitative strategy is one that can be profitable **over the long term**, rather than one that makes money on every trade.

Murphy's Law: Anything that can go wrong will go wrong.

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Strive for Excellence

6 Strive for Excellence, Not Perfection

- There is no perfect strategy in the world, so don't strive for perfection in quantitative investing; instead, aim for excellence.
- The uncertainty in quantitative strategies **should come solely from the market, not from human intervention.** If the strategy's performance is affected by even the slightest deviation due to human factors, it is the responsibility of the person, not the strategy itself.

> *If these strategies were easy to live with, they'd likely be far more susceptible to being arbitraged away.* — Cliff Asness, AQR Capital

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7 The “Magic” Formula

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- Cognitive Bias
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- Tendency for Overfitting
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- The “Magic” Formula

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- Single Strategy:

$$\alpha > 0 \textbf{ per trade} + \textbf{Law of Large Numbers}.$$

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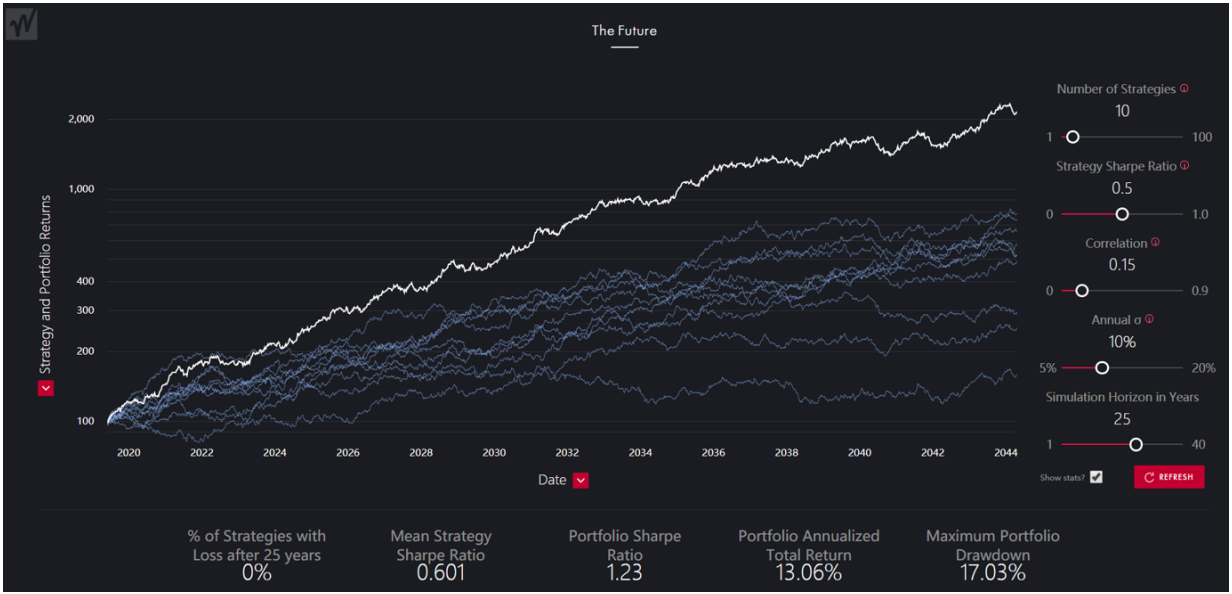
- Single Strategy:

$$\alpha > 0 \textbf{ per trade} + \textbf{Law of Large Numbers}.$$

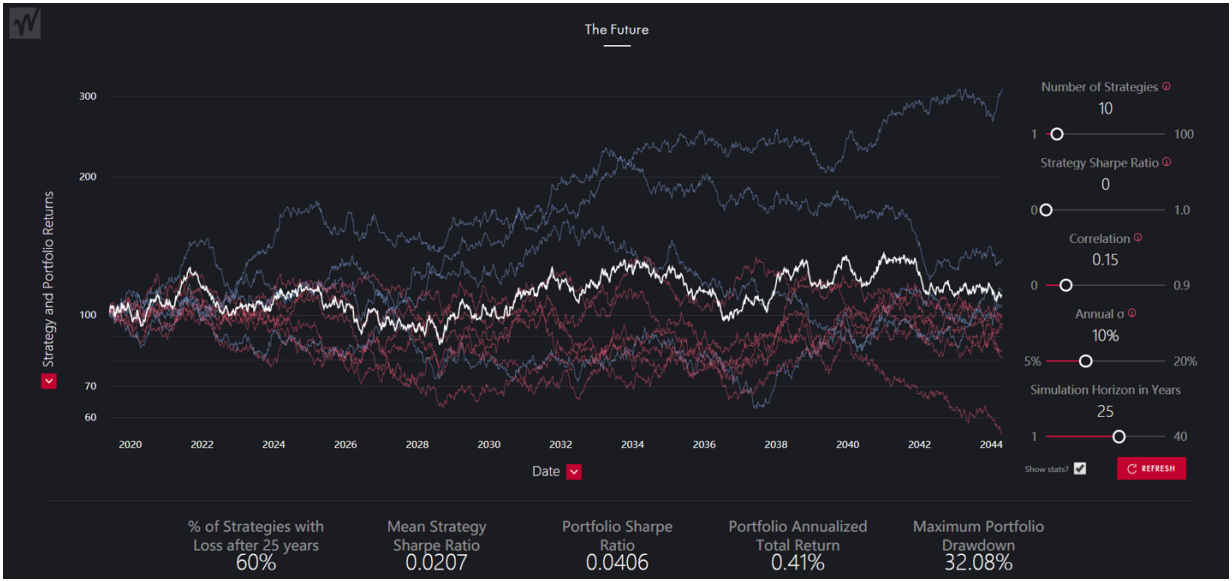
- Investment:

$$\alpha > 0 \textbf{ per trade} + \textbf{Law of Large Numbers} + \textbf{Low Correction}.$$

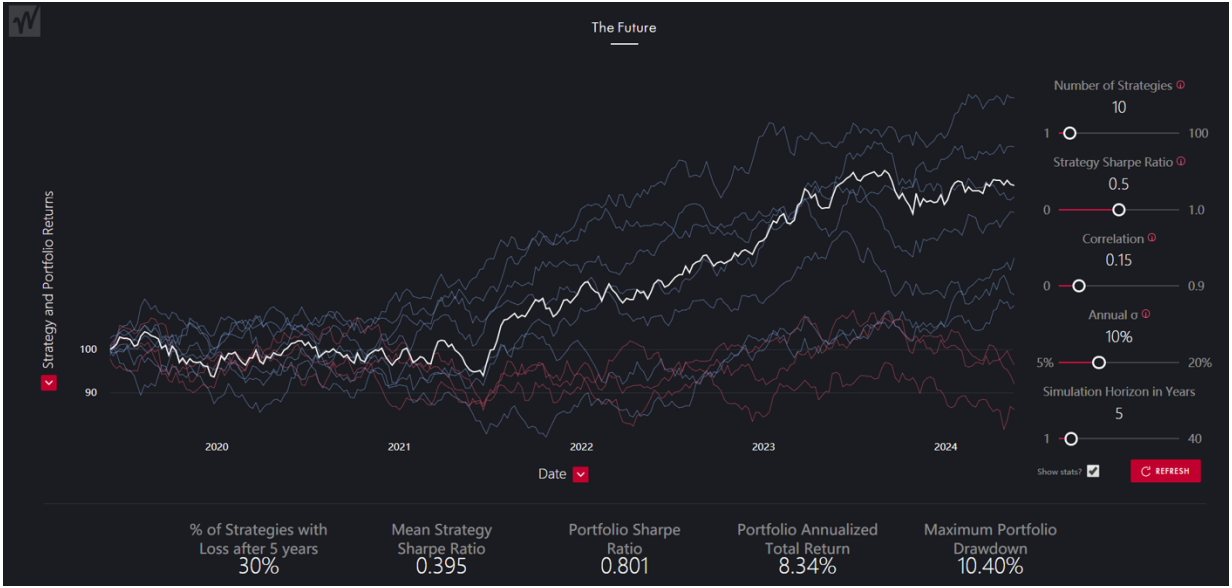
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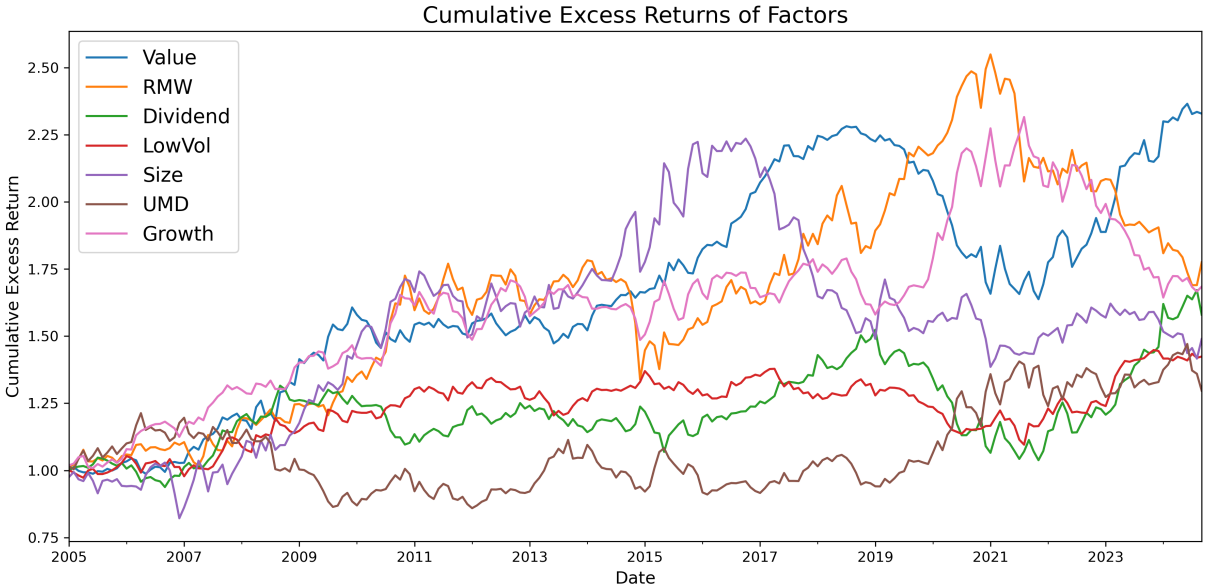
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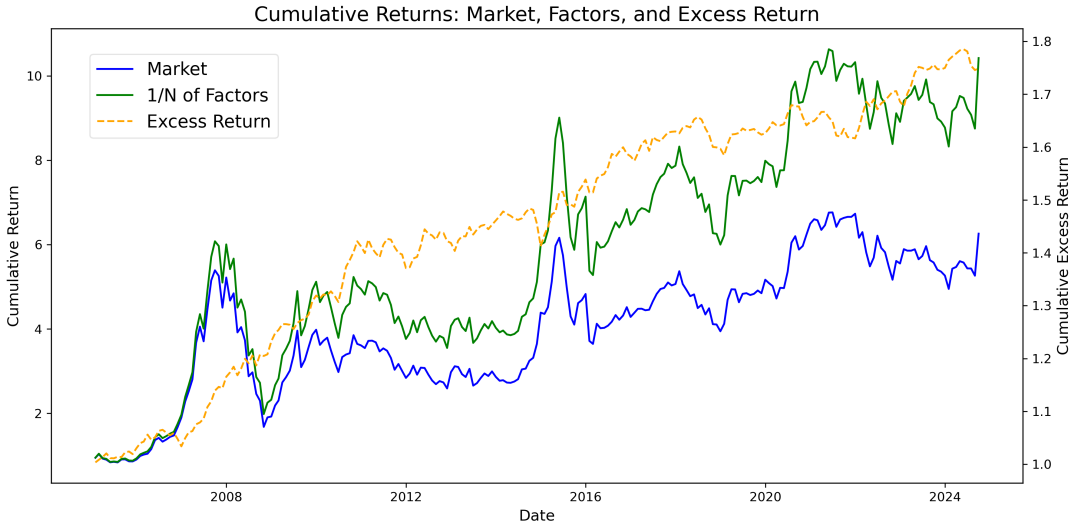


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	Value	RMW	Dividend	Low Vol	Size	UMD	Growth
Panel A: Monthly Average Excess Return (%)							
	0.38	0.28	0.22	0.17	0.23	0.15	0.25
Panel B: Correlation Coefficient							
Value	1.00	−0.29	0.54	0.41	0.05	−0.35	−0.27
RMW		1.00	−0.46	−0.19	0.14	0.32	0.43
Dividend			1.00	0.27	−0.38	−0.49	−0.63
Low Vol				1.00	−0.03	−0.39	−0.30
Size					1.00	0.02	0.28
UMD						1.00	0.55
Growth							1.00

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- The 1/N factor portfolio's monthly excess return is 0.20% (t-statistic = 4.11).



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Questions?

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